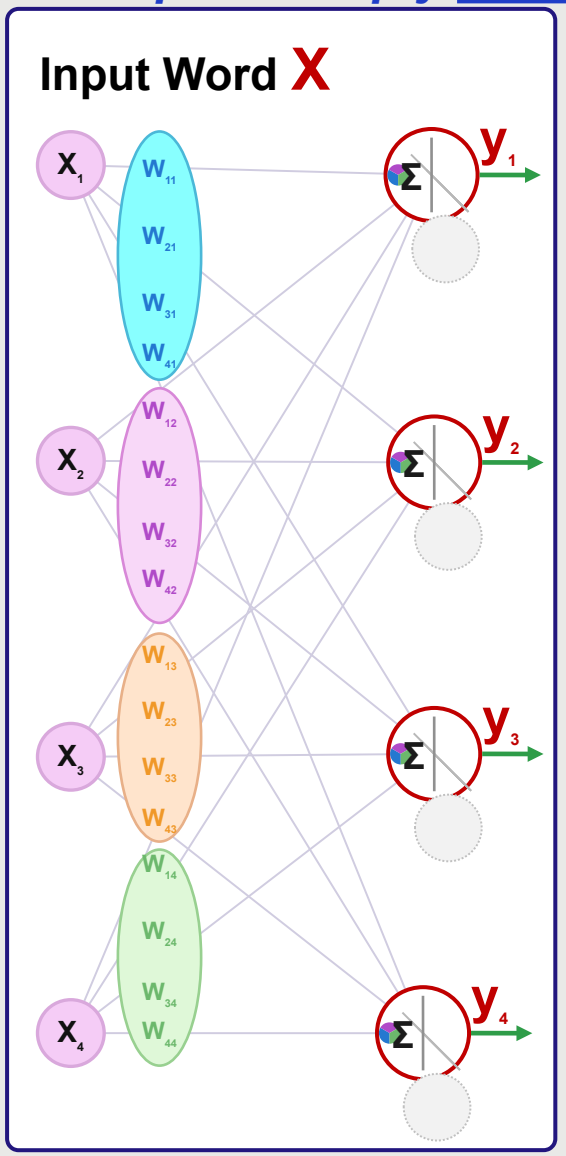
Using FF NN for Multiplication of ONE Vector by a Matrix

Example: Multiply Vector of Current Word X by W_Q (Query) {Remember: W_Q is Squared Matrix}



Y^T

$[\ y_1\ y_2\ y_3\ y_4\ ]$ =Sigmoid

IF No Non-linearity

Y^T

$[\ y_1\ y_2\ y_3\ y_4\ ]$ =Linear

$[\ x_1\ x_2\ x_3\ x_4\ ]$ X

$\begin{bmatrix} W_{11} & W_{21} & W_{31} & W_{41} \\ W_{12} & W_{22} & W_{32} & W_{42} \\ W_{13} & W_{23} & W_{33} & W_{43} \\ W_{14} & W_{24} & W_{34} & W_{44} \end{bmatrix}$ + $\begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix}$

IF No Bias

$[\ x_1\ x_2\ x_3\ x_4\ ]$ X

$\begin{bmatrix} W_{11} & W_{21} & W_{31} & W_{41} \\ W_{12} & W_{22} & W_{32} & W_{42} \\ W_{13} & W_{23} & W_{33} & W_{43} \\ W_{14} & W_{24} & W_{34} & W_{44} \end{bmatrix}$ + $\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$